Calculus Review, Part II

Bridging Mathematics, Fall 2025

1. Evaluate the following indefinite integrals.

   (a) $\displaystyle\int \frac{1+x}{1+x^2}\,dx$

   (b) $\displaystyle\int x^2 \ln x\,dx$

   (c) $\displaystyle\int \tan^3 x\,dx$

(a) We split the integral into sums:

$$\int \frac{1+x}{1+x^2}\,dx = \int \frac{1}{1+x^2}\,dx + \int \frac{x}{1+x^2}\,dx.$$

To evaluate the first integral, we recall that

$$\frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}.$$

To evaluate the second integral, we employ a substitution: let $u = 1+x^2$, with $du = 2x\,dx$. Then,

$$\int \frac{x}{1+x^2}\,dx = \int \frac{1}{2}\cdot\frac{1}{u}\,du = \frac{1}{2}\ln|u| + c.$$

Since $u = 1+x^2$, we find that

$$\int \frac{1+x}{1+x^2}\,dx = \tan^{-1} x + \frac{1}{2}\ln\left|1+x^2\right| + c.$$

(b) We integrate by parts: let $u = \ln x$ and $dv = x^2\,dx$. Then,

$$u = \ln x \implies du = \frac{1}{x}\,dx, \qquad dv = x^2\,dx \implies v = \frac{x^3}{3}.$$

Then,

$$\begin{aligned}\int x^2 \ln x\,dx &= \ln x\cdot\frac{x^3}{3} - \int \frac{x^3}{3}\cdot\frac{1}{x}\,dx \\ &= \frac{1}{3}x^3\ln x - \frac{1}{3}\int x^2\,dx \\ &= \frac{1}{3}x^3\ln x - \frac{1}{3}\cdot\frac{x^3}{3} + c \\ &= \frac{1}{3}x^3\ln x - \frac{1}{9}x^3 + c.\end{aligned}$$

(c) Observe that we may rewrite the integrand as

$$\tan^3 x = \tan x \tan^2 x.$$

In particular, since $\tan^2 x + 1 = \sec^2 x$, we have

$$\int \tan^3 x\,dx = \int \tan x\left(\sec^2 x - 1\right)\,dx = \int \tan x \sec^2 x\,dx - \int \tan x\,dx.$$

To evaluate the first integral, we let $u = \tan x$, with $du = \sec^2 x\,dx$. Then,

$$\int \tan x \sec^2 x\,dx = \int u\,du = \frac{u^2}{2} + c.$$

Substituting $u = \tan x$ back, we conclude that

$$\int \tan^3 x\,dx = \frac{1}{2}\tan^2 x - \ln|\sec x| + c.$$

2. Evaluate the following definite integrals.

(a) $\displaystyle\int_{-2}^{2} f(x)\, dx$, where

$$f(x) = \begin{cases} 2 & \text{for } x \leq 0, \\ 4 - x^2 & \text{for } x > 0. \end{cases}$$

(b) $\displaystyle\int_{0}^{\pi} e^{\cos\theta} \sin\theta \, d\theta$

(c) $\displaystyle\int_{0}^{\infty} \frac{1}{x^2 + 5x + 6}\, dx$

(a) We begin by splitting the integral:

$$\int_{-2}^{2} f(x)\, dx = \int_{-2}^{0} 2\, dx + \int_{0}^{2} 4 - x^2\, dx.$$

We can thus proceed to integrate term-by-term:

$$\int_{-2}^{0} 2\, dx = 2x|_{-2}^{0} = 2(0) - 2(-2) = 4,$$

$$\int_{0}^{2} 4 - x^2\, dx = 4x - \frac{x^3}{3}\bigg|_{0}^{2} = \left(4 \cdot 2 - \frac{2^3}{3}\right) - \left(4 \cdot 0 - \frac{0^3}{3}\right) = \frac{16}{3}$$

Hence, we find that

$$\int_{-2}^{2} f(x)\, dx = 4 + \frac{16}{3} = \frac{28}{3}.$$

(b) We perform a substitution: let $u = \cos\theta$, with $du = -\sin\theta\, d\theta$. Note that the bounds of integration go from $\cos 0 = 1$ to $\cos\pi = -1$. We thus have

$$\int_{0}^{\pi} e^{\cos\theta} \sin\theta\, d\theta = \int_{1}^{-1} -e^u\, du = e^u|_{-1}^{1} = e - \frac{1}{e}.$$

(c) To begin, observe that we may factor the denominator

$$x^2 + 5x + 6 = (x+2)(x+3).$$

We peform a partial fraction decomposition on the integrand:

$$\frac{A}{x+2} + \frac{B}{x+3} = \frac{A(x+3) + B(x+2)}{(x+2)(x+3)} = \frac{(A+B)x + 3A + 2B}{x^2 + 5x + 6}.$$

Comparing coefficients, we requrie that

$$A + B = 0, \quad 3A + 2B = 1.$$

We thus find that $A = 1$, and $B = -1$. Hence,

$$\begin{aligned} \int_{0}^{\infty} \frac{1}{x^2 + 5x + 6}\, dx &= \int_{0}^{\infty} \frac{1}{x+2} - \frac{1}{x+3}\, dx \\ &= \lim_{t\to\infty} \left[\ln|x+2| - \ln|x+3|\right]_0^t \\ &= \lim_{t\to\infty} \left[\ln\left|\frac{t+2}{t+3}\right| - \ln\left(\frac{2}{3}\right)\right]. \end{aligned}$$

We evaluate the first term of the limit:

$$\lim_{t\to\infty} \ln\left|\frac{t+2}{t+3} \cdot \frac{1/t}{1/t}\right| = \lim_{t\to\infty} \ln\left|\frac{1+2/t}{1+3/t}\right| = \ln 1 = 0.$$

Hence, we conclude that

$$\int_{0}^{\infty} \frac{1}{x^2 + 5x + 6}\, dx = 0 - \ln\left(\frac{2}{3}\right) = \ln\left(\frac{3}{2}\right).$$

3. Determine the values of p for which the following integral converges:

$$\int_0^1 x^p \ln x \, dx.$$

To begin, note that the integral is improper since the integrand is undefined when $x = 0$. We first find the antiderivative of $x^p \ln x$. If $p = -1$, observe that we can integrate by substitution: letting $u = \ln x$ with $du = 1/x \, dx$,

$$\int \frac{1}{x} \ln x \, dx = \int u \, du = \frac{u^2}{2} + c = \frac{(\ln x)^2}{2} + c.$$

Then, the improper integral is given by

$$\lim_{t \to 0} \frac{(\ln x)^2}{2}\bigg|_t^1 = \lim_{t \to 0} \left(\frac{(\ln 1)^2}{2} - \frac{(\ln t)^2}{2} \right) = -\lim_{t \to 0} \frac{(\ln t)^2}{2} \to -\infty,$$

and the integral diverges. When $p \neq -1$, we integrate by parts—let $u = \ln x$ and $dv = x^p \, dx$:

$$u = \ln x \implies du = \frac{1}{x} \, dx, \qquad dv = x^p \, dx \implies v = \frac{x^{p+1}}{p+1}.$$

Then, we have

$$\begin{aligned} \int x^p \ln x \, dx &= \ln x \cdot \frac{x^{p+1}}{p+1} - \int \frac{x^{p+1}}{p+1} \cdot \frac{1}{x} \, dx \\ &= \ln x \cdot \frac{x^{p+1}}{p+1} - \frac{1}{p+1} \int x^p \, dx \\ &= \ln x \cdot \frac{x^{p+1}}{p+1} - \frac{x^{p+1}}{(p+1)^2} + c. \end{aligned}$$

To evaluate the improper integral, we instead consider a limit:

$$\begin{aligned} \lim_{t \to 0} \int_t^1 x^p \ln x \, dx &= \lim_{t \to 0} \left[\ln x \cdot \frac{x^{p+1}}{p+1} - \frac{x^{p+1}}{(p+1)^2}\bigg|_t^1 \right] \\ &= \lim_{t \to 0} \left[-\frac{1}{(p+1)^2} - \left(\ln t \cdot \frac{t^{p+1}}{p+1} - \frac{t^{p+1}}{(p+1)^2} \right) \right] \\ &= \lim_{t \to 0} \left[-\frac{1}{(p+1)^2} - \frac{1}{p+1} \left(\ln t \cdot t^{p+1} \right) + \frac{t^{p+1}}{(p+1)^2} \right] \\ &= -\frac{1}{(p+1)^2} - \frac{1}{p+1} \lim_{t \to 0} \left[\ln t \cdot t^{p+1} \right] \end{aligned}$$

We now evaluate the final limit: observe that when $p > -1$, we can apply l'Hôpital's rule:

$$\lim_{t \to 0} \left[\ln t \cdot t^{p+1} \right] = \lim_{t \to 0} \frac{\ln t}{t^{-p-1}} = \lim_{t \to 0} \frac{1/t}{(-p-1)\, t^{-p-2}} = \lim_{t \to 0} -\frac{1}{p+1} t^{p+1} = 0.$$

When $p < -1$, however, we have

$$\lim_{t \to 0} \left[\ln t \cdot t^{p+1} \right] = \lim_{t \to 0} \frac{\ln t}{t^{-(p+1)}} \to -\infty,$$

and the limit does not exist. Hence, we require that $p > -1$ for the integral to converge.

4. Use the substitution $u = 1/x$ to evaluate

$$\int_0^\infty \frac{\ln x}{1+x^2}\,dx.$$

Let $u = 1/x$, and so $x = 1/u$, with $dx = -1/u^2\,du$. Then, the integral becomes

$$\int_{x=0}^{x=\infty} \frac{\ln(1/u)}{1+1/u^2}\cdot -\frac{1}{u^2}\,du = \int_\infty^0 \frac{\ln u}{u^2+1}\,du = -\int_0^\infty \frac{\ln u}{1+u^2}\,du.$$

Note that this is simply the negative of the original integral that we started with! We thus have the following equality:

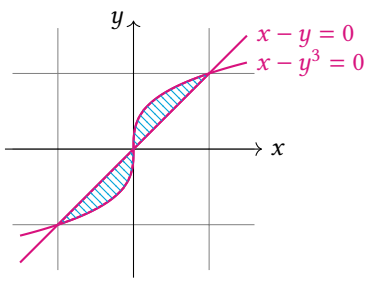
$$\int_0^\infty \frac{\ln x}{1+x^2}\,dx = -\int_0^\infty \frac{\ln x}{1+x^2}\,dx,$$

and such an equality between two numbers is possible if and only if they were both equal to 0. Hence,

$$\int_0^\infty \frac{\ln x}{1+x^2}\,dx = 0.$$

5. Find the area of the region enclosed by the curve $x - y^3 = 0$ and the line $x - y = 0$.

We begin by sketching the region.



We determine the points of intersection: we require that $x = y^3$ and $x = y$, and so

$$y = y^3 \quad\Longrightarrow\quad 0 = y^3 - y = y\left(y^2-1\right) = y\,(y+1)\,(y-1)\,.$$

Thus, the two graphs intersect at $(0,0)$, $(-1,-1)$, and $(1,1)$. Note that the graphs are symmetric about the origin; thus, it suffices to consider the area of the shaded region in the first quadrant—the area of the region in the third quadrant will be the same. We compute the area of the enclosed region in the first quadrant: the region is bounded above by $y = \sqrt[3]{x}$, and below by $y = x$. Thus,

$$\int_0^1 \sqrt[3]{x} - x\,dx = \frac{3}{4}x^{4/3} - \frac{1}{2}x^2\Big|_{x=0}^{x=1} = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}.$$

Thus, the total area of the shaded region must be $2 \times 1/4 = 1/2$.

Alternatively, we could have considered the region as bounded on the left by $x = y^3$ and on the right by $x = y$:

$$\int_0^1 y - y^3 = \frac{y^2}{2} - \frac{y^4}{4}\Big|_{y=0}^{y=1} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4},$$

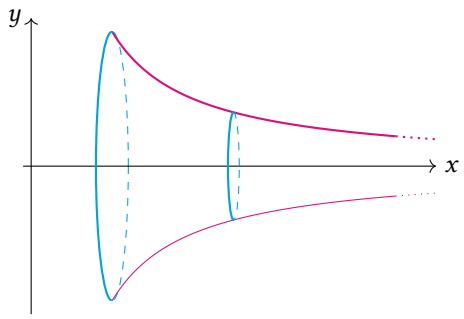
and we arrive at the same result.

6. Consider the region

$$R = \{(x, y) \mid x \geq 1,\ 0 \leq y \leq 1/x\}.$$

Suppose that R is rotated about the x-axis. Show that the surface area of the solid formed is infinite.

We sketch the region R.



Note that

$$\frac{dy}{dx} = \frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}.$$

The surface area of the region is given by

$$\int_1^\infty 2\pi\left(\frac{1}{x}\right)\sqrt{1+\left(-\frac{1}{x^2}\right)^2}\,dx = \int_1^\infty 2\pi\left(\frac{1}{x}\right)\sqrt{1+\frac{1}{x^4}}\,dx.$$

This is not an easy integral to compute. Note, however, that for all x,

$$\sqrt{1+\frac{1}{x^4}} > \sqrt{1} = 1.$$

Multiplying both sides by $2\pi/x$ with $x > 0$,

$$2\pi\left(\frac{1}{x}\right)\sqrt{1+\frac{1}{x^4}} > 2\pi\left(\frac{1}{x}\right).$$

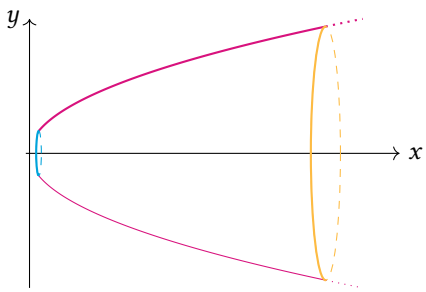
In particular, we can conclude that

$$2\pi\int_1^\infty \left(\frac{1}{x}\right)\sqrt{1+\frac{1}{x^4}}\,dx > 2\pi\int_1^\infty \frac{1}{x}\,dx.$$

Since the integral on the right-hand side diverges to infinity, the integral on the left-hand side must diverge as well: that is, the surface area of the region must be infinite.

7. A cup is obtained by rotating the parabola $y^2 = 4x$ about the x-axis. Suppose that the radius of the base of the cup is 1 cm, and the cup has volume 32π cm^3. Determine the diameter of the of the rim of the cup.

To begin, the base of the cup occurs when the radius $y = 1$; that is, $x = 1/4$. We sketch the graph of $y^2 = 4x$ (or $y = 2\sqrt{x}$) for $x \geq 1/4$.



We first find where the rim of the cup occurs: its radius occurs at the value of $x = b$ such that the volume of the solid of revolution from $x = 1/4$ to $x = b$ is 32π: that is,

$$\int_{1/4}^{b} \pi \left(2\sqrt{x}\right)^2 \, dx = 32\pi.$$

We evaluate the integral:

$$\int_{1/4}^{b} \pi \left(2\sqrt{x}\right)^2 \, dx = 4\pi \int_{1/4}^{b} x \, dx = 4\pi \left[\frac{x^2}{2}\right]_{x=1/4}^{x=b} = 2\pi \left(b^2 - \frac{1}{16}\right).$$

We thus require that

$$2\pi \left(b^2 - \frac{1}{16}\right) = 32\pi \quad \Longrightarrow \quad b^2 - \frac{1}{16} = 16 \quad \Longrightarrow \quad b = \sqrt{\frac{257}{16}}.$$

The radius of the cup at $x = b$ is thus

$$y = (4b)^{1/2} = \left(4\sqrt{\frac{257}{16}}\right)^{1/2} = 257^{1/4}.$$

Hence, the diameter of the rim is $2 \times 257^{1/4} \approx 8.00780$ cm.

8. Determine a particular solution to each of the following first-order differential equations.

   (a) $xy^2y' = x+1$, where $y(1) = 1$
   (b) $x\ln x = y\left(1+\sqrt{3+y^2}\right)y'$, where $y(1) = 1$
   (c) $y'\tan x = a+y$, where $y(\pi/3) = a$

We perform a separation of variables for each differential equation.

(a) We have

$$xy^2\frac{dy}{dx} = x+1 \quad\Longrightarrow\quad \int y^2\,dy = \int \frac{x+1}{x}\,dx.$$

Integrating both sides, we have

$$\frac{y^3}{3} = x+\ln|x|+c.$$

We determine the arbitrary constant c: since $y(1) = 1$,

$$\frac{1}{3} = 1+\ln|1|+c \quad\Longrightarrow\quad c = \frac{1}{3}-1 = -\frac{2}{3}.$$

Hence, a particular solution is given by

$$\frac{y^3}{3} = x+\ln|x|-\frac{2}{3}.$$

(b) We have

$$x\ln x = y\left(1+\sqrt{3+y^2}\right)\frac{dy}{dx} \quad\Longrightarrow\quad \int x\ln x\,dx = \int y\left(1+\sqrt{3+y^2}\right)dy.$$

We evaluate the integral on the left by parts. Let $u = \ln x$ with $du = 1/x\,dx$, and $dv = x\,dx$ with $v = x^2/2$:

$$\int x\ln x\,dx = \ln x\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot\frac{1}{x}\,dx = \frac{1}{2}x^2\ln x-\frac{x^2}{4}+c.$$

We now expand the integral on the right:

$$\int y+y\sqrt{3+y^2}\,dy = \frac{y^2}{2}+\int y\sqrt{3+y^2}\,dy.$$

We integrate by substitution: let $u = 3+y^2$, with $du = 2y\,dy$. Then,

$$\begin{aligned}\frac{y^2}{2}+\int y\sqrt{3+y^2}\,dy &= \frac{y^2}{2}+\int\frac{1}{2}\sqrt{u}\,dy\\ &= \frac{y^2}{2}+\frac{1}{2}\cdot\frac{u^{3/2}}{3/2}+c\\ &= \frac{y^2}{2}+\frac{1}{3}\left(3+y^2\right)^{3/2}+c.\end{aligned}$$

Hence, a general solution to the differential equation is given by

$$\frac{1}{2}x^2\ln x-\frac{x^2}{4} = \frac{y^2}{2}+\frac{1}{3}\left(3+y^2\right)^{3/2}+c.$$

We determine the value of the arbitrary constant c: since $y(1) = 1$, we have

$$\frac{1}{2}\ln 1-\frac{1}{4} = \frac{1}{2}+\frac{1}{3}(3+1)^{3/2}+c \quad\Longrightarrow\quad c = -\frac{1}{4}-\frac{1}{2}-\frac{8}{3} = -\frac{41}{12}.$$

Hence, the desired particular solution is

$$\frac{1}{2}x^2\ln x-\frac{x^2}{4} = \frac{y^2}{2}+\frac{1}{3}\left(3+y^2\right)^{3/2}-\frac{41}{12}.$$

(c) We have

$$\frac{dy}{dx}\tan x = a + y \quad \Longrightarrow \quad \int \frac{1}{a+y}\,dy = \int \cot x\,dx.$$

Directly integrating, we have

$$\ln|a+y| = \ln|\sin x| + c \quad \Longrightarrow \quad a + y = C\sin x.$$

Since $y(\pi/3) = a$, we find that

$$a + a = C\sin(\pi/3) \quad \Longrightarrow \quad C = 4a/\sqrt{3}.$$

Hence, the desired particular solution is given by

$$y = \frac{4a}{\sqrt{3}}\sin x - a.$$

9. A certain drug is injected into a patient at a constant rate r. In the blood, the drug is converted into other substances and excreted at a rate proportional to its concentration at that time. We can thus use a differential equation to model the concetration $x(t)$ of the drug in the patient's bloodstream:

$$\frac{dx}{dt} = r - kx, \text{ for some } k > 0.$$

(a) Suppose that, at time $t = 0$, the concentration of the drug is x_0. Solve the differential equation to determine the concentration of the drug at any time t.

(b) Suppose that $x_0 < r/k$. Find the limit of $x(t)$ as $t \to \infty$. What does this value represent?

We perform a separation of variables:

$$\int \frac{1}{r-kx}\,dx = \int 1\,dt.$$

To evaluate the integral on the left, we perform a substitution: let $u = r - kx$, $du = -k\,dx$. Then,

$$\int \frac{1}{r-kx}\,dx = \int -\frac{1}{k}\cdot\frac{1}{u}\,du = -\frac{1}{k}\ln|u| + c.$$

Since $u = r - kx$, we have the general solution

$$-\frac{1}{k}\ln|r-kx| = t + c \quad \Longrightarrow \quad \ln|r-kx| = -kt + d.$$

Exponentiating both sides yields

$$r - kx = Ce^{-kt} \quad \Longrightarrow \quad x = \frac{r}{k} - De^{-kt}.$$

We determine the constant D: we have $x(0) = x_0$, and so

$$x_0 = \frac{r}{k} - D \quad \Longrightarrow \quad D = \frac{r}{k} - x_0.$$

Hence, the concentration of the drug at any time t is given by

$$x(t) = \frac{r}{k} - \left(\frac{r}{k} - x_0\right)e^{-kt}.$$

Observe that as $t \to \infty$, the second term in $x(t)$ approaches 0, and so $x(t) \to r/k$. This represents the concentration of the drug left in the body after a very long time.

10. A single case of a newly discovered flu virus has been found in one of the dormitories at a local university. In order to curb the spread of the disease on campus, the affected dormitory, which houses exactly N students, goes into lockdown, such that no one is allowed to enter or leave the building.

    Let $y(t)$ be the number of students in the dormitory who have been exposed to the virus at time t days. Suppose that the rate dy/dt at which the disease spreads is equal to the product between the number of students who have contracted the virus and the number of students who have not yet been exposed to it.

    (a) Write down a differential equation, alongside any initial conditions, that models $y(t)$, the number of students who have contracted the virus at time t days.

    (b) Determine the time it takes for all but one student in the dormitory to contract the virus.

(a) $\dfrac{dy}{dt} = y(N-y)$, with $y(0) = 1$.

(b) We perform a separation of variables:

$$\int \frac{1}{y(N-y)}\,dy = \int dt \quad \Longrightarrow \quad \frac{1}{N}\ln\left|\frac{y}{y-N}\right| = t + c.$$

Since $y(0) = 1$, we have

$$c = \frac{1}{N}\ln\left|\frac{1}{1-N}\right| \quad \Longrightarrow \quad t = \frac{1}{N}\ln\left|\frac{y}{y-N}\right| - \frac{1}{N}\ln\left|\frac{1}{1-N}\right|.$$

Thus, the time it takes for all but one student to contract the virus is

$$t^* = \frac{1}{N}\ln\left|\frac{N-1}{N-1-N}\right| - \frac{1}{N}\ln\left|\frac{1}{1-N}\right| = \frac{2}{N}\ln|1-N|.$$